Light

GCSE Computer Science - Boolean Logic (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

An AND gate requires which condition to output 1?

- **A.** At least one input must be 0
- **B.** Inputs must be different
- **C.** At least one input must be 1
- **D.** All inputs must be 1

Q2 (1 mark)

An OR gate outputs 0 only when:

- **A.** All inputs are 0

- ○ **B.** All inputs are 1
- ○ **C.** Inputs are different
- ○ **D.** At least one input is 1

Q3 (1 mark)

How many inputs does a NOT gate have?

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** 2
- ○ **D.** 3

Q4 (1 mark)

XOR stands for:

- ○ **A.** Extra OR
- ○ **B.** Extended OR
- ○ **C.** Exclusive OR
- ○ **D.** External OR

Q5 (1 mark)

A logic circuit has 3 inputs. How many rows are in its truth table?

- A. 3 rows
- B. 6 rows
- C. 9 rows
- D. 8 rows

Q6 (1 mark)

What is the output of an AND gate when A = 0 and B = 0?

- A. 0
- B. 1
- C. Undefined
- D. Either 0 or 1

Q7 (1 mark)

What is the output of an OR gate when A = 1 and B = 1?

- A. 0
- B. 2
- C. 1
- D. Undefined

Q8 (1 mark)

What is the output of an XOR gate when A = 0 and B = 0?

- ○ **A.** 1
- ○ **B.** 2
- ○ **C.** Undefined
- ○ **D.** 0

Q9 (1 mark)

If the input to a NOT gate is 0, what is the output?

- ○ **A.** 1
- ○ **B.** 0
- ○ **C.** -1
- ○ **D.** Undefined

Q10 (1 mark)

In Boolean algebra, how is NOT A typically written?

- ○ **A.** A + 1
- ○ **B.** A - 1
- ○ **C.** $\bar{A}$ (A with an overbar)

- **D.** A × 0

Q11 (1 mark)

A logic circuit produces 8 different input combinations in its truth table. How many inputs does this circuit have?

- **A.** 2 inputs
- **B.** 8 inputs
- **C.** 4 inputs
- **D.** 3 inputs

Q12 (1 mark)

Which gate would be used to check if a value is NOT equal to 1?

- **A.** AND gate
- **B.** OR gate
- **C.** NOT gate
- **D.** XOR gate

Q13 (2 marks)

In a circuit, inputs A and B go into an AND gate. The AND output and input C go into an OR gate. If A = 1, B = 0, and C = 1, what is the final output?

- A. 0
- B. 1
- C. Cannot be determined
- D. Depends on gate order

Q14 (2 marks)

Input A passes through a NOT gate. The NOT output and input B go into an OR gate. If A = 0 and B = 0, what is the output?

- A. 1
- B. 0
- C. Undefined
- D. Both inputs needed

Q15 (2 marks)

Evaluate the Boolean expression A + B when A = 1 and B = 1.

- A. 0
- B. 2
- C. 1
- D. 11

Q16 (2 marks)

What is the value of A.B.C when A = 1, B = 1, and C = 0?

- **A.** 1
- **B.** 0
- **C.** 3
- **D.** Undefined

Q17 (2 marks)

Two sensors each output 0 or 1. A warning light should turn on when the sensors give different readings. Which gate should be used?

- **A.** AND gate
- **B.** OR gate
- **C.** NOT gate
- **D.** XOR gate

Q18 (2 marks)

Evaluate $A \oplus B$ when A = 0 and B = 1.

- **A.** 0
- **B.** 1
- **C.** -1

- **D.** 01

Q19 (2 marks)

In a NOT gate truth table, how many rows have an output of 1?

- **A.** 1 row
- **B.** 0 rows
- **C.** 2 rows
- **D.** All rows

Q20 (2 marks)

A circuit has inputs A and B. Both go into an AND gate, then the result goes through a NOT gate. If A = 1 and B = 1, what is the final output?

- **A.** 1
- **B.** 0
- **C.** Depends on NOT placement
- **D.** Cannot determine

Q21 (2 marks)

Evaluate the expression $\bar{A}.B$ when A = 0 and B = 1.

- **A.** 0

- **B.** 1
- **C.** Undefined
- **D.** -1

Q22 (2 marks)

For which input combination do AND and OR gates give the same output?

- **A.** A = 0, B = 1 only
- **B.** A = 1, B = 0 only
- **C.** A = 0, B = 0 and A = 1, B = 1
- **D.** A = 0, B = 1 and A = 1, B = 0

Q23 (2 marks)

Inputs A and B go into an XOR gate. The XOR output and input C go into an AND gate. If A = 1, B = 1, and C = 1, what is the final output?

- **A.** 1
- **B.** 0
- **C.** Undefined
- **D.** Depends on C

Q24 (3 marks)

A circuit has the expression A + (B.C). In the truth table for inputs A, B, C, how many rows have output 1?

- **A.** 4 rows
- **B.** 5 rows
- **C.** 6 rows
- **D.** 7 rows

Q25 (3 marks)

Which Boolean expression represents: 'A and B go into an OR gate, then the result is ANDed with C'?

- **A.** A.B + C
- **B.** A + B.C
- **C.** (A + B).C
- **D.** A.(B + C)

Q26 (3 marks)

Evaluate the expression $(A + B)\overline{\ }$ when A = 0 and B = 0.

- **A.** 0
- **B.** 1
- **C.** 2

- **D.** Undefined

Q27 (3 marks)

A circuit has inputs A, B, and C. A and B feed into an XOR gate. The XOR output and C feed into an OR gate. For which input combination is the final output 0?

- **A.** A = 0, B = 0, C = 0
- **B.** A = 0, B = 1, C = 0
- **C.** A = 1, B = 0, C = 0
- **D.** A = 1, B = 1, C = 1

Q28 (3 marks)

A 2-input truth table shows output Q. Q = 1 for rows (0,1) and (1,0) only. Q = 0 for rows (0,0) and (1,1). Which gate produces this pattern?

- **A.** AND gate
- **B.** OR gate
- **C.** NOT gate
- **D.** XOR gate

Q29 (3 marks)

A heating system should activate (output 1) when the temperature sensor (T) shows cold AND either the timer (M) is on OR the manual override (O) is pressed. Which expression represents this?

- **A.** T + M + O
- **B.** T.M.O
- **C.** T.(M + O)
- **D.** (T.M) + O

Q30 (3 marks)

A truth table for 3 inputs (A, B, C) shows output Q = 1 only for row (1,1,1). Which expression matches this?

- **A.** A + B + C
- **B.** A.B.C
- **C.** A ⊕ B ⊕ C
- **D.** (A + B).C

Total: **55** marks • Time: **30 min**

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